For a quarter wave plate,

$$d = (2m+1)\frac{\lambda_0}{4(n_e - n_o)};\ m = 0,1,2...$$

Thus, if in Fig. 22.26, the calcite QWP is replaced by a quartz QWP, the beam emerging from the QWP will be right circularly polarized.

Example 22.7 We consider a left circularly polarized beam ($\lambda_0 = 5893\ \text{Å} = 0.5893\ \mu\text{m}$) incident normally on a calcite crystal (with its optic axis cut parallel to the surface) of thickness 0.005141 mm. The electric field for the incident left circularly polarized beam at $z = 0$ can be written as

$$E_x = E_1 \sin\omega t\ ;\ E_y = E_1 \cos\omega t \qquad (22.55)$$

Now

$$\theta = \frac{(n_o - n_e)d \times 2\pi}{\lambda_0} = \frac{0.17195 \times 5.141 \times 10^{-6} \times 2\pi}{5.893 \times 10^{-7}} \approx 3\pi$$

Thus the emergent wave will be [cf. Eq. (22.49)]

$$E_x = E_1 \sin(\omega t - 3\pi) = -E_1 \sin\omega t\ ;\ E_y = E_1 \cos\omega t$$

which represents a right circularly polarized beam.

Example 22.8 We next consider a left circularly polarized beam ($\lambda_0 = 5893\ \text{Å}$) is incident on a quartz crystal (with its optic axis cut parallel to the surface) of thickness 0.022 mm. We assume n_o and n_e to be 1.54425 and 1.55336 respectively. The electric field for the incident left circularly polarized beam at $z = 0$ would be given by Eq. (22.55). Further,

$$\theta' = (n_e - n_o)\frac{2\pi}{\lambda_0}d = 2\pi\ \frac{0.00911 \times 2.2 \times 10^{-5}}{5.893 \times 10^{-7}} \approx 0.68\pi$$

Thus the emergent beam will be

$$E_x = E_1 \sin(\omega t + 0.68\pi)\ ;\ E_y = E_1 \cos\omega t$$

which will represent a right elliptically polarized light.

22.7 ANALYSIS OF POLARIZED LIGHT LO 5

In the earlier sections we have seen that a plane wave can be characterized by different states of polarizations, which may be anyone of the following:

(a) linearly polarized
(b) circularly polarized
(c) elliptically polarized
(d) unpolarized
(e) mixture of linearly polarized and unpolarized
(f) mixture of circularly polarized and unpolarized
(g) mixture of elliptically polarized and unpolarized light

To the naked eye all the states of polarizations will appear to be the same. In this section, we will discuss the procedure for determining the state of polarization of a light beam.

If we introduce a polaroid in the path of the beam and rotate it about the direction of propagation, then either of the following three possibilities can occur:

(i) If there is complete extinction at two positions of the polarizer, then the beam is linearly polarized.
(ii) If there is no variation of intensity, then the beam is either unpolarized or circularly polarized or a mixture of unpolarized and circularly polarized light. We now put a quarter wave plate on the path of the beam followed by the rotating polaroid. If there is no variation of intensity then the incident beam is unpolarized. If there is complete extinction at two positions, then the beam is circularly polarized (this is due to the fact that a quarter wave plate will transform a circularly polarized light into a linearly polarized light). If there is a variation of intensity (without complete extinction) then the beam is a mixture of unpolarized and circularly polarized light.
(iii) If there is a variation of intensity (without complete extinction) then the beam is either elliptically polarized or a mixture of linearly polarized and unpolarized or a mixture of elliptically polarized and unpolarized light. We now put a quarter wave plate in front of the polaroid with its optic axis parallel to the pass-axis of the polaroid at the position of maximum intensity. The elliptically polarized light will transform to a linearly polarized light. Thus, if one obtains two positions of the polaroid where complete extinction occurs, then the original beam is elliptically polarized. If complete extinction does not occur, and the position of maximum intensity occurs at the same orientation as before, the beam is a mixture of unpolarized and linearly polarized light. Finally, if the position of maximum intensity occurs at a different orientation of the polaroid, the beam is a mixture of elliptically polarized and unpolarized light.

22.8 OPTICAL ACTIVITY LO 6

When a linearly polarized light beam propagates through an "optically active" medium like sugar solution then—as the beam propagates—its plane of polarization rotates. This rotation is directly proportional to the distance traversed by

the beam and also to the concentration of sugar in the solution. Indeed, by measuring the angle by which the plane of polarization is rotated, one can accurately determine the concentration of sugar in the solution.

The rotation of the plane of polarization is due to the fact that the "modes" of the optically active substance are left circularly polarized (LCP) and right circularly polarized (RCP) which propagate with slightly different velocities (see Sec. 22.16). By "modes" we imply that if an LCP light beam is incident on the substance then it will propagate as an LCP beam; similarly, an RCP light beam will propagate as an RCP beam but with a slightly different velocity. On the other hand, if a linearly polarized light beam is incident, then we must express the linear polarization as a superposition of an RCP and an LCP beam and then consider the independent propagation of the two beams. We illustrate through an example.

We consider an RCP beam propagating in the +z direction

$$\left.\begin{aligned} E_x^R &= E_0 \cos(k_R z - \omega t) \\ E_y^R &= -E_0 \sin(k_R z - \omega t) \end{aligned}\right\} \qquad \textbf{(22.56)}$$

where $k_R = \dfrac{\omega}{c} n_R$ and the superscript (and the subscript) R signify that we are considering an RCP beam. Similarly, an LCP beam (of the same amplitude) propagating in the $+z$ direction can be described by the following equations:

$$\left.\begin{aligned} E_x^L &= E_0 \cos(k_L z - \omega t) \\ E_y^L &= E_0 \sin(k_L z - \omega t) \end{aligned}\right\} \qquad \textbf{(22.57)}$$

where $k_L = \dfrac{\omega}{c} n_L$; n_R and n_L are the refractive indices corresponding to the RCP and LCP beams respectively. If we assume the simultaneous propagation of the two beams then the x and y components of the resultant fields would be given by the following equations:

$$E_x = E_0 \left[\cos(k_R z - \omega t) + \cos(k_L z - \omega t)\right]$$

or

$$E_x = 2E_0 \cos\left[\frac{1}{2}(k_L - k_R)\, z\right] \cos[\omega t - \theta(z)]$$

Similarly

$$E_y = 2E_0 \sin\left[\frac{1}{2}(k_L - k_R)\, z\right] \cos[\omega t - \theta(z)]$$

where

$$\theta(z) = \frac{1}{2}(k_L + k_R) z$$

Thus the resultant wave is *always* linearly polarized with the plane of polarization rotating with z. If the direction of the oscillating electric vector makes an angle ϕ with the x-axis then [see Fig. 22.28]:

$$\phi(z) = \frac{1}{2}(k_L - k_R)\, z = \frac{\pi}{\lambda}(n_L - n_R)\, z = \frac{\omega}{2c}(n_L - n_R)\, z \qquad \textbf{(22.58)}$$

where λ is the free space wavelength. Now, if

$n_L > n_R \;\Leftrightarrow$ the optically active substance is said to be right-handed or dextro-rotatory

$n_L < n_R \;\Leftrightarrow$ the optically active substance is said to be left-handed or laevo-rotatory

For example, for turpentine, $\phi = +37$ for $z = 10$ cm.

As mentioned earlier, we observe optical activity even in a sugar solution, and this is due to the helical structure of sugar molecules. The method of determining the concentration of sugar solutions by measuring the rotation of the plane of polarization is a widely used method in industry. It may be noted that if $n_L = n_R$ (as is indeed the case in an isotropic substance) then $\phi(z) = 0$ and a linearly polarized beam remains linearly polarized along the same direction. Optical activity is also exhibited in crystals. For example, for a linearly polarized light propagating along the optic axis of a quartz crystal*, the plane of polarization gets rotated. Indeed

$$|n_L - n_R| \approx 7\times10^{-5} \quad \Rightarrow \quad \phi \approx \frac{7}{60}\pi = 21^\circ \text{ for } z = 0.1 \text{ cm};$$

$\lambda_0 = 0.6\ \mu\text{m} = 6000$ Å

22.9 CHANGE IN THE SOP (STATE OF POLARIZATION) OF A LIGHT BEAM PROPAGATING THROUGH AN ELLIPTIC CORE SINGLE-MODE OPTICAL FIBER LO 7

A very interesting phenomenon is the propagation of polarized light through an elliptic core optical fiber. We will have a brief discussion on optical fibers in Chapter 29; it will suffice here to say that in an ordinary optical fiber we have a cylindrical core (of circular cross-section) cladded with a

* When a wave propagates along the optic axis of a quartz crystal it is strictly speaking, not like calcite. The modes are *not* linearly polarized; they are RCP and LCP propagating with slightly different velocities.

medium of slightly lower refractive index. The guidance of the light beam takes place through the phenomenon of total internal reflection (see Fig. 29.1). Because of the circular symmetry of the problem, the incident beam can have any state of polarization* which will be maintained as the beam propagates through the fiber. Now, if we have an elliptic core fiber [see Fig. 22.29(a)] then the "modes" of the fiber are (approximately) x and y polarized; i.e., if an x-polarized beam is incident it will propagate without any change in the state of polarization with a certain phase velocity ω/β_x. Similarly, a y-polarized beam will propagate as a y-polarized beam with velocity ω/β_y. Now, let a circularly polarized beam be incident on the input face of the fiber at $z = 0$. Then we must resolve the incident beam into x and y polarized beams propagating with slightly different velocities. Thus

$$\mathbf{E}(x, y, z) = \psi(x, y)\,[\hat{\mathbf{x}} \cos(\beta_x z - \omega t) + \hat{\mathbf{y}} \sin(\beta_y z - \omega t)] \tag{22.59}$$

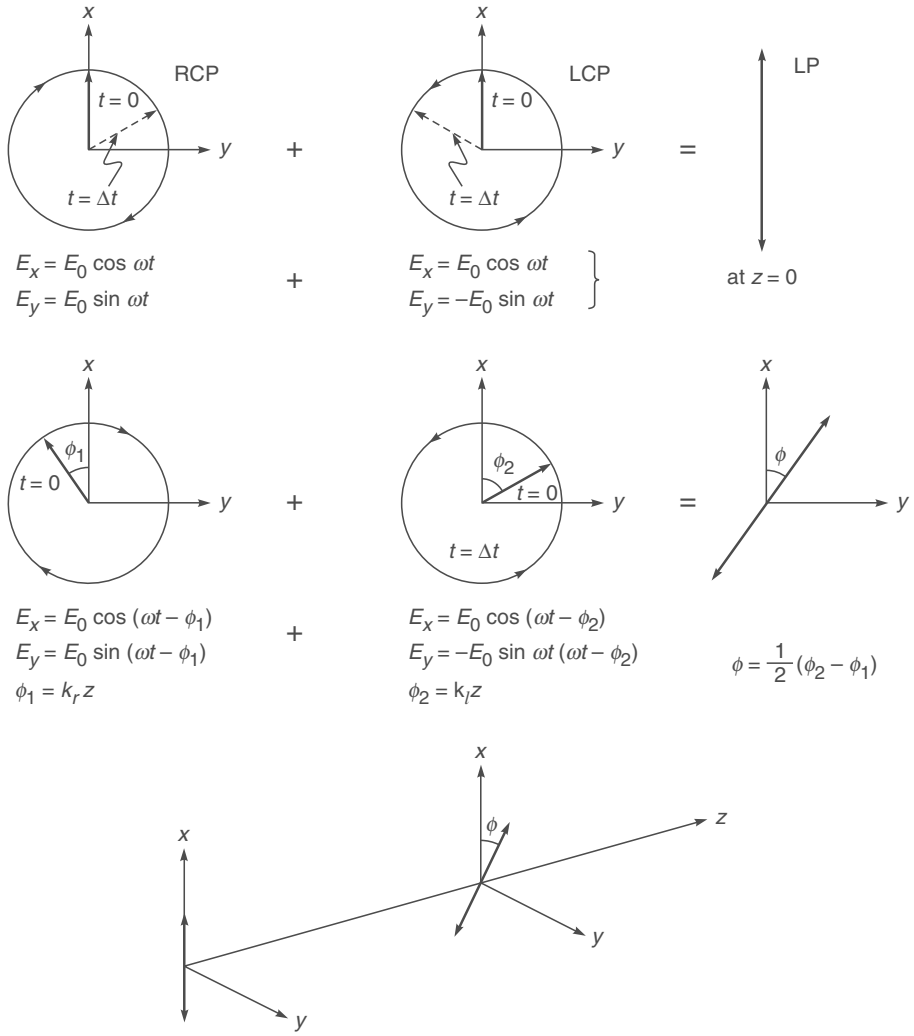


Fig. 22.28 The "clockwise" rotation of a linearly polarized wave as it propagates through a "right-handed" optically active medium.

* We are considering here a single mode fiber so that no matter what the incident transverse field distribution is, it soon "settles down" to the transverse field distribution of the fundamental mode which propagates with the velocity ω/β_0. This velocity is independent of the SOP of the incident beam.

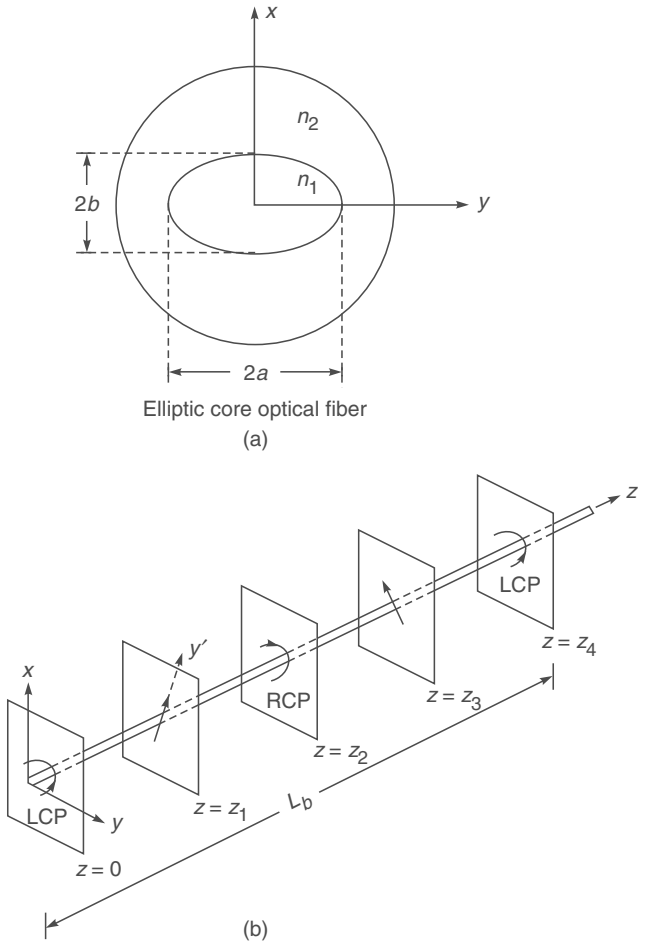


Fig. 22.29 (a) The transverse cross-section of an elliptic core fiber; the "modes" are (approximately) x-polarized and y-polarized. (b) Propagation of a left-circularly polarized beam incident on an elliptic core fiber. If we view along the y - axis then dark spots will be observed at $z = z_1, 5z_1, 9z_1, \ldots$

where $\psi(x, y)$ is the transverse field distribution of the fundamental mode which is assumed to be (approximately) the same for both x- and y-polarizations. It may be readily seen that if $\beta_x = \beta_y$, as is indeed true for circular core fibers, the beam will remain circularly polarized for all values of z. Now, at $z = 0$,

$$E_x = \psi(x, y) \cos \omega t$$
$$E_y = -\psi(x, y) \sin \omega t \qquad \textbf{(22.60)}$$

which represents a left-circularly polarized wave [see Fig. 22.29 (b)]. For

$$z = z_1 = \frac{\pi}{2(\beta_y - \beta_x)} \qquad \textbf{(22.61)}$$

i.e., for $\beta_y z_1 = \beta_x z_1 + \pi/2$,

$$E_x = \psi(x, y) \cos(\phi_1 - \omega t) = +\psi(x, y) \cos(\omega t - \phi_1)$$

$$E_y = \psi(x, y) \sin\left(\phi_1 + \frac{\pi}{2} - \omega t\right) = +\psi(x, y) \cos(\omega t - \phi_1)$$

where

$$\phi_1 = \beta_x z_1$$

which represents a linearly polarized wave [see Fig. 22.29(b)]; we assume the direction of the **E**-vector to be along the y' axis. Similarly, at

$$z = z_2 = \frac{\pi}{(\beta_y - \beta_x)} = 2 z_1$$

$$E_x = \psi(x, y) \cos(\phi_2 - \omega t) = \psi(x, y) \cos(\omega t - \phi_2)$$
$$E_y = \psi(x, y) \sin(\phi_2 + \pi - \omega t) = \psi(x, y) \sin(\omega t - \phi_2) \qquad \textbf{(22.62)}$$

where

$$\phi_2 = \beta_x z_2$$

and the wave will be right-circularly polarized [see Fig. 22.29(b)]. At

$$z = z_3 = \frac{3\pi}{2(\beta_x - \beta_y)} = 3 z_1$$

we will have

$$E_x = \psi(x, y) \cos(\phi_3 - \omega t) = \psi(x, y) \cos(\omega t - \phi_3)$$

$$E_y = \psi(x, y) \sin\left(\phi_3 + \frac{3\pi}{2} - \omega t\right) = -\psi(x, y) \cos(\omega t - \phi_3)$$

where

$$\phi_3 = \beta_x z_3$$

Thus the wave would again be linearly polarized but now the direction of the oscillating electric field will be at right angles to the field at $z = z_1$. In a similar manner, we can easily continue to determine the SOP of the propagating beam. Thus at $z = 5z_1, 9z_1, 13z_1, \ldots$ the SOP will be the same as at $z = z_1$ and at $z = 7z_1, 11z_1, 15z_1 \ldots$ the SOP will be the same as at $z = 3z_1$. Similarly at $z = 4z_1, 8z_1, 12z_1 \ldots$ the beam will be LCP and at $z = 2z_1, 6z_1, 10z_1 \ldots$ the beam will be RCP.

Now, let the fiber be rotated in such a way that the y' axis is along the vertical line (the x' and the z-axes are assumed to lie in the horizontal plane). Thus if we put our eyes vertically above the fiber and view vertically down then the regions $z = z_1, 5z_1, 9z_1 \ldots$ will appear dark (see Fig. 22.30). This is